

IMPERIAL

jeff update

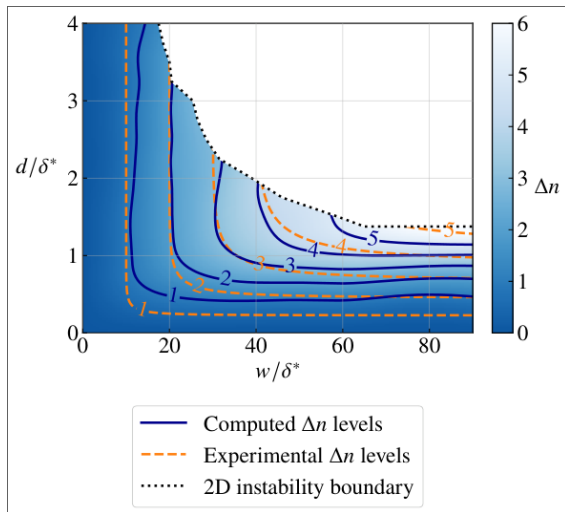
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Δn -factors

Difference in computation.

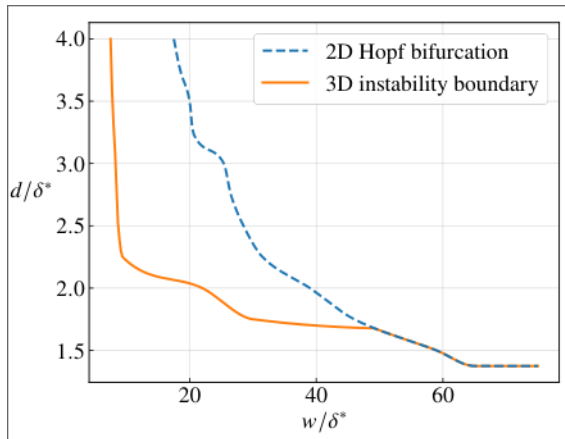
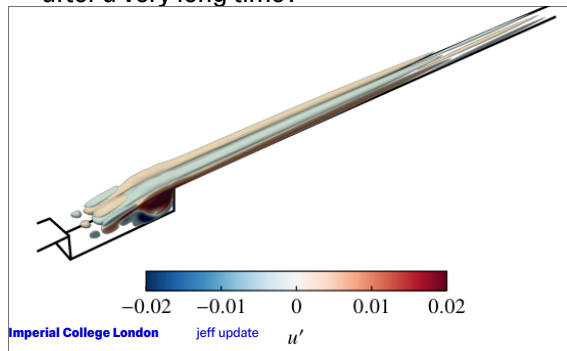
- Jeff (??): First computes the n -factor $n(x)$ curve for the flat plate as a function of x . Then, observes the transition location x_t^{flat} and x_t^{gap} and sets $n_{\text{flat}} := n(x_t^{\text{flat}})$ and $n_{\text{gap}} := n(x_t^{\text{gap}})$. Of course since $x_t^{\text{gap}} < x_t^{\text{flat}}$, then $n_{\text{gap}} < n_{\text{flat}}$ because n is monotonic.
- Victor: Use the definition of $n(x)$ as $\log(A(x)/A_0)$ for both the flat plate and the gap case in different computations. So $n_{\text{flat}} = \log(A_{\text{flat}}(x_t^{\text{flat}})/A_0)$ and $n_{\text{gap}} = \log(A_{\text{gap}}(x_t^{\text{gap}})/A_0)$. Since $A_{\text{gap}}(x) > A_{\text{flat}}(x)$ then $n_{\text{gap}} > n_{\text{flat}}$.

Same physics? idk



3d stuff

- Between the 2d and 3d bifurcations, it doesn't seem to be transition, because the streaks in the BL are very weak. But of course the integration of time is finite and the velocity evolution inside the gap is not clear (quite chaotic). Could it be that the transition come after a very long time?



Ma number

- Ma number generally destabilizes the system.
- However Ma = 0.8 seems to behave slightly different than 0.6 and 0.4. Why???
- We also ran incompressible 2d simulations at Re = 2000 (instead of 1000) and the Hopf bifurcation shifts to smaller widths.

